

A high-speed micro-ring modulator for next generation energy-efficient optical networks beyond 100 Gbaud

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Abstract: We demonstrate a silicon micro-ring modulator supporting 128 Gb/s NRZ modulation with SNR=5.2, ER=3.8dB, 0.8 Vpp drive swing, and 5.3 fJ/bit power consumption. We have also achieved 192 Gb/s PAM-4 modulation with TDECQ of 2.5dB.

1. Introduction

Silicon optical modulators provide low-cost solutions for high-speed optical communications and interconnects. Such modulators are generally based on Mach-Zehnder interferometers that can support high baud rate applications [1]. However, high-speed silicon micro-ring modulators (MRMs) have recently emerged as an attractive alternative to MZMs due to their compact size and low power consumption [2-4]. In a 2020 demonstration, a silicon micro-ring modulator based 1.6 Tb/s optical engine was co-packaged with a 12.8Tb/s Ethernet switch and shown to interoperate with a standard DR4 pluggable module [5]. A photonic chip with such high bandwidth density was only achievable by using compact micro-ring modulators. In this paper, we report a high-speed silicon micro-ring modulator with a record-breaking speed of 128 Gbaud on-off keying (OOK) modulation. We have achieved a clear open eye diagram with signal to noise ratio (SNR) of 5.2 and extinction ratio (ER) of 3.8 dB measured directly with a digital communication analyzer (DCA) without any offline digital signal processing (DSP). The drive voltage swing was 0.8 Vpp and the MRM consumed less than 5.3 fJ/bit power. This is a significant milestone for next generation energy-efficient high-speed optical communication networks beyond 100 Gbaud with low-cost silicon optical modulators.

2. Modulator Design and Characterization

Figure 1 shows a schematic of the silicon photonic MRM design. The micro-ring was formed by a silicon rib waveguide with height of 300-nm, width of 400-nm, and slab thickness of 100 nm. The MRM has a radius of 6 μm , resulting in a Free Spectral Range (FSR) of 11.3 nm at 1310 nm. The loaded quality factor of the MRM was measured to be 4200. The electro-optic (EO) phase efficiency ($V_\pi \cdot L_\pi$) of the PN junction of the MRM as a function of the reverse bias is shown in Fig. 2. The measured $V_\pi \cdot L_\pi$ is 0.53 V·cm at a reverse bias voltage of 3 V and 0.42 V·cm at 0 V bias. Compared to our previous work [2], the junction was redesigned to support higher data rate of >100 Gbaud and lower heater voltage of $V_\pi < 1.5\text{V}$ for thermo-optic phase tuning.

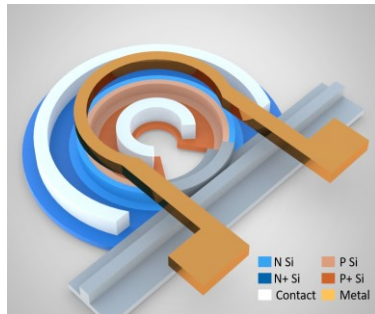


Fig. 1. A schematic of the silicon photonic MRM with metal heater.

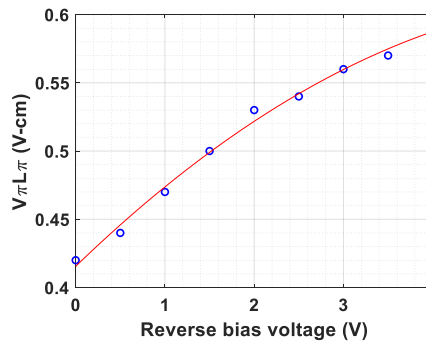


Fig. 2. Modulation efficiency as a function of reverse bias.

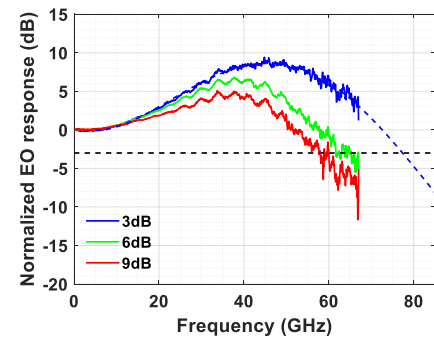


Fig. 3. Measured EO response (S21) at reverse bias of 3V for 3 wavelength detuning levels of 3, 6 and 9 dB.

The electrical S11 parameter of the MRM was measured at a reverse bias voltage of 3 V using a vector network analyzer. The responses of the cables and the probe were de-embedded through standard calibration. The extracted resistance and capacitance of the PN junction were 26 Ω and 33 fF, respectively. The EO response (S21) was measured using a Lightwave Component Analyzer (LCA) at reverse bias of 3V for three laser wavelength detuning points corresponding to 3, 6 and 9 dB down from the off-resonance transmission maximum power level. As shown in Fig. 3, the measured 3-dB RF modulation bandwidths were 77, 63, and 58 GHz, correspondingly.

EO bandwidths

Figures 4 and 5 show the **measured electrical and optical eye diagrams**. We used an arbitrary waveform generator followed by a RF amplifier and a 50- Ω terminated probe to drive the MRM. The optical input was provided by an O-band tunable laser and the optical output from the MRM was amplified by an optical amplifier and sent to the DCA. Fig. 4(a) shows the 128 Gb/s electrical driving signal with SNR of 4.6 and RF swing of 0.8 V_{pp} . We obtained an open eye with **SNR of 2.7 and ER of 3.9 dB** at a wavelength detuning point of -9 dB, as shown in Fig. 4(b), without any equalization. By applying a 7-tap **feed forward equalization (FFE)**, the SNR can be improved to 5.2, Fig. 4(c). Based on the eye SNR (Q-factor), we estimated the bit error ratio (BER) would be 3.4×10^{-3} at the receiver without any equalization [6]. The power consumption of the MRM was calculated to be 5.3 fJ/bit **based on the drive voltage and its capacitance**. We also drove the MRM with a 96 Gbaud or 192 Gb/s PAM-4 electrical signal with a voltage swing of 1.6 V_{pp}, Fig. 5(a). The directly measured PAM-4 optical eye diagram showed ER of 3.5 with no FFE. For better linearity we operated the MRM at a wavelength detuning point of -6 dB. By applying a 5-tap FFE, we obtained an improved PAM-4 eye with a **TDECQ** of 2.5 dB and an outer ER of 3.4 dB, as shown in Fig. 5(c). We believe the obtained results can be further improved by using a dedicated modulator driver with matched impedance and higher bandwidth.

a DSP technique applied at the receiver to improve signal quality, compensate for (ISI) and channel distortions that blur the eye diagram at very high data rates.

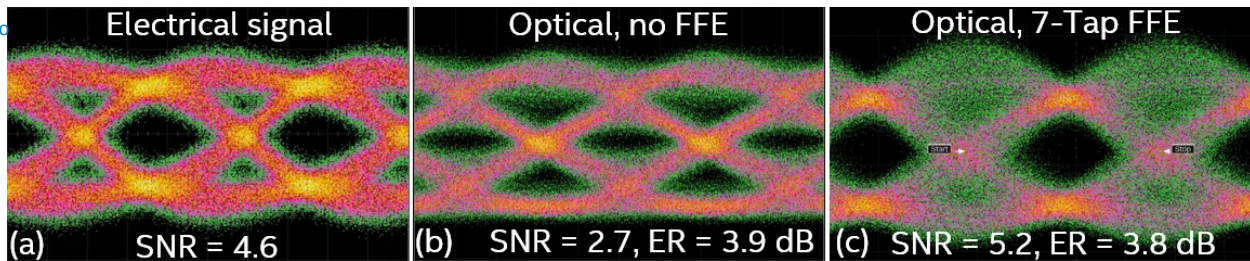


Fig. 4 NRZ Eye diagrams at 128 Gbaud. (a) electrical input; (b) optical eye without FFE; and (c) optical eye with 7-tap FFE.

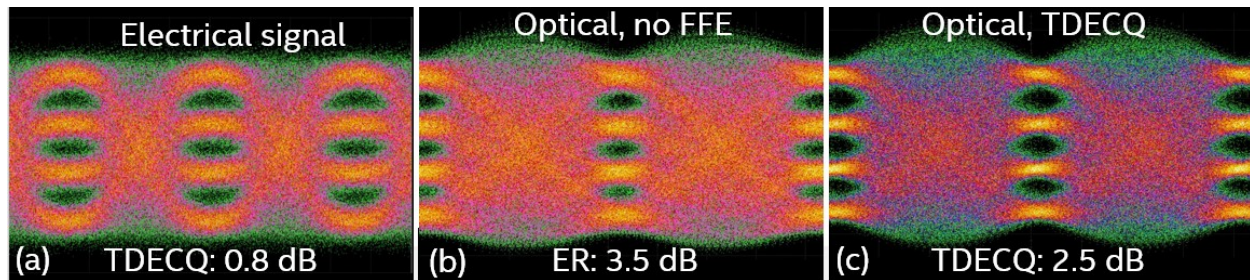


Fig. 5 PAM-4 Eye diagrams at 96 Gbaud or 192 Gb/s (a) electrical input; (b) optical w/o equalizer; and (c) optical eye with 5-tap FFE.

4. Conclusion

We demonstrated, to the best of our knowledge, the first silicon modulator operating at 128 Gb/s NRZ modulation consuming 5.3 fJ/bit **without any DSP**. We obtained eye SNR of 5.2 with a 7-tap receiver FFE, supporting hard-decision forward error correction with a threshold of $\text{BER} < 3.8 \times 10^{-3}$ [7]. We also demonstrated 192 Gb/s PAM-4 modulation with a TDECQ of 2.5 dB without any transmitter pre-equalization. We believe that the silicon MRM is a promising candidate for next generation short-reach optical networks beyond 100 Gbaud per channel with compact size and low power consumption [8].

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4. References

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